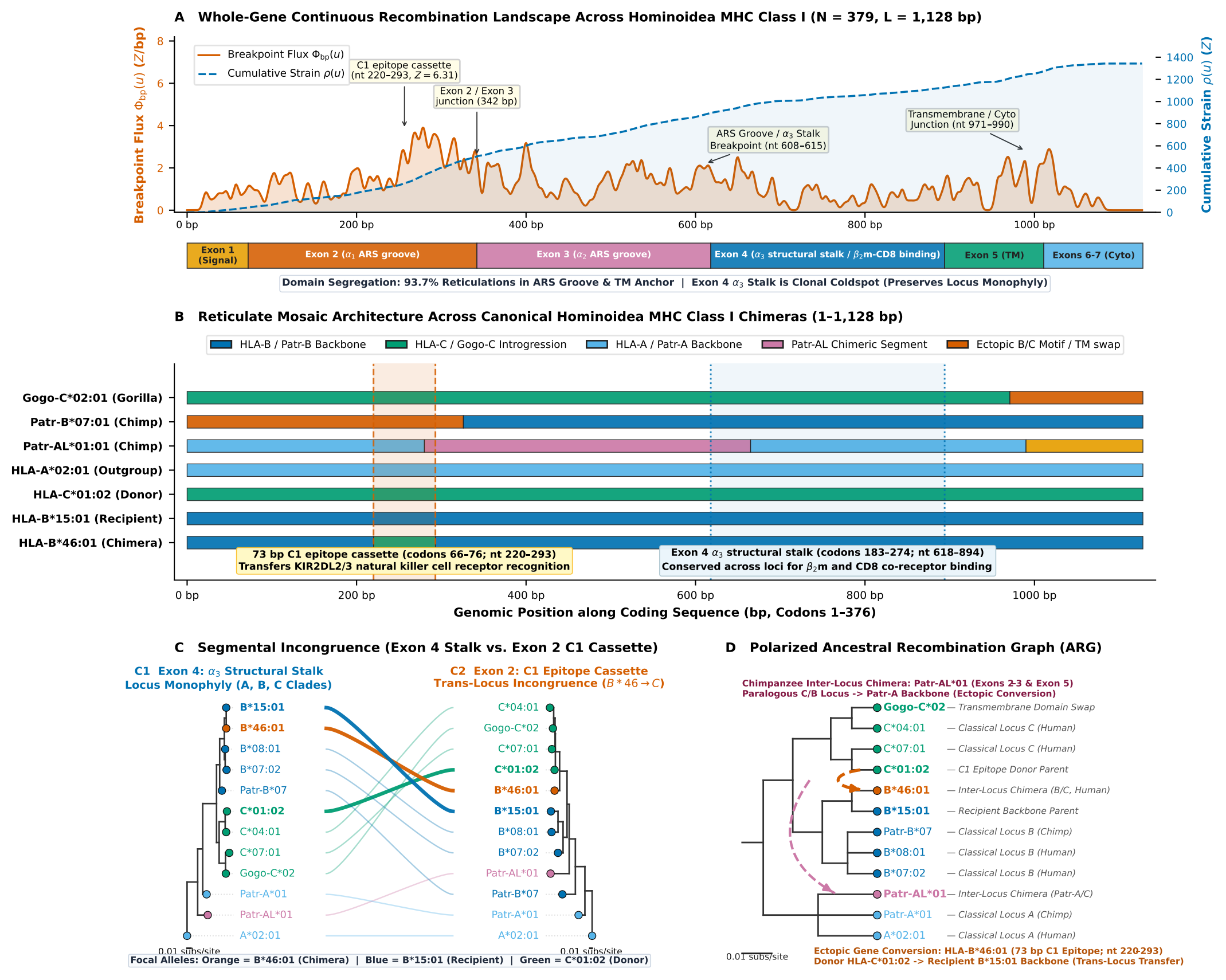

A Whole-Gene Continuous Recombination Landscape Across Hominoidea MHC Class I (N = 379, L = 1,128 bp)
Breakpoint Flux $\Phi_{bp}(u)$ (Z/bp)
Cumulative Strain $\rho(u)$ (Z)
Breakpoint Flux $\Phi_{bp}(u)$
Cumulative Strain $\rho(u)$
C1 epitope cassette (nt 220–293, Z = 6.31)
Exon 2 / Exon 3 junction (342 bp)
ARS Groove / α_3 Stalk Breakpoint (nt 608–615)
Transmembrane / Cyto Junction (nt 971–990)
0 bp
200 bp
400 bp
600 bp
800 bp
1000 bp
Exon 1 (Signal)
Exon 2 (α_1 ARS groove)
Exon 3 (α_2 ARS groove)
Exon 4 (α_3 structural stalk / β_2m-CD8 binding)
Exon 5 (TM)
Exons 6-7 (Cyto)
Domain Segregation: 93.7% Reticulations in ARS Groove & TM Anchor | Exon 4 α_3 Stalk is Clonal Coldspot (Preserves Locus Monophyly)
B Reticulate Mosaic Architecture Across Canonical Hominoidea MHC Class I Chimeras (1–1,128 bp)
HLA-B / Patr-B Backbone
HLA-C / Gogo-C Introgression
HLA-A / Patr-A Backbone
Patr-AL Chimeric Segment
Ectopic B/C Motif / TM swap
Gogo-C*02:01 (Gorilla)
Patr-B*07:01 (Chimp)
Patr-AL*01:01 (Chimp)
HLA-A*02:01 (Outgroup)
HLA-C*01:02 (Donor)
HLA-B*15:01 (Recipient)
HLA-B*46:01 (Chimera)
73 bp C1 epitope cassette (codons 66–76; nt 220–293)
Transfers KIR2DL2/3 natural killer cell receptor recognition
Exon 4 α_3 structural stalk (codons 183–274; nt 618–894)
Conserved across loci for β_2m and CD8 co-receptor binding
Genomic Position along Coding Sequence (bp, Codons 1–376)
C Segmental Incongruence (Exon 4 Stalk vs. Exon 2 C1 Cassette)
C1 Exon 4: α_3 Structural Stalk
Locus Monophyly (A, B, C Clades)
C2 Exon 2: C1 Epitope Cassette
Trans-Locus Incongruence ($B*46 \rightarrow C$)
B*15:01
B*46:01
B*08:01
B*07:02
Patr-B*07
C*01:02
C*04:01
C*07:01
Gogo-C*02
Patr-A*01
Patr-AL*01
A*02:01
0.01 subs/site
Focal Alleles: Orange = B*46:01 (Chimera) | Blue = B*15:01 (Recipient) | Green = C*01:02 (Donor)
D Polarized Ancestral Recombination Graph (ARG)
Chimpanzee Inter-Locus Chimera: Patr-AL*01 (Exons 2-3 & Exon 5)
Paralogous C/B Locus -> Patr-A Backbone (Ectopic Conversion)
Gogo-C*02 — Transmembrane Domain Swap
C*04:01 — Classical Locus C (Human)
C*07:01 — Classical Locus C (Human)
C*01:02 — C1 Epitope Donor Parent
B*46:01 — Inter-Locus Chimera (B/C, Human)
B*15:01 — Recipient Backbone Parent
Patr-B*07 — Classical Locus B (Chimp)
B*08:01 — Classical Locus B (Human)
B*07:02 — Classical Locus B (Human)
Patr-AL*01 — Inter-Locus Chimera (Patr-A/C)
Patr-A*01 — Classical Locus A (Chimp)
A*02:01 — Classical Locus A (Human)
0.01 subs/site
Ectopic Gene Conversion: HLA-B*46:01 (73 bp C1 Epitope; nt 220-293)
Donor HLA-C*01:02 -> Recipient B*15:01 Backbone (Trans-Locus Transfer)